DOSSIER 4 : CODE R - ANALYSE STATISTIQUE

Projet : Facteurs Prédictifs de la Réussite Étudiante

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1 Chargement des bibliothèques et des données

```r
if(!require(nnet)) install.packages("nnet")
if(!require(tidyverse)) install.packages("tidyverse")
if(!require(aod)) install.packages("aod")
if(!require(caret)) install.packages("caret")

library(nnet)
library(tidyverse)
library(aod)
library(caret)

# Chargement du dataset (bas  sur le fichier dataset.csv)
data <- read.csv("dataset.csv", sep = ",", check.names = FALSE)
```

2 Nettoyage et préparation des données

```r
# Nettoyage des noms de colonnes pour la syntaxe R
colnames(data) <- make.names(colnames(data))

# Traitement de la variable cible 'Target'
data$Target <- as.factor(data$Target)

# MODIFICATION : D finition de la classe de r f rence sur 'Dropout'
data$Target <- relevel(data$Target, ref = "Dropout")

# Conversion des variables explicatives cl s en facteurs
categorical_vars <- c("Gender", "Scholarship.holder", "Debtor", "
    Tuition.fees.up.to.date")
data[categorical_vars] <- lapply(data[categorical_vars], as.factor)
```

3 Exécution du modèle logistique multinomial

```r
# Mod le incluant les variables acad miques , socio-d mographiques et
    macro
model <- multinom(Target ~ Gender +
                    Scholarship.holder +
                    Age.at.enrollment +
```

```
                    Curricular.units.1st.sem..grade. +
                    Curricular.units.2nd.sem..grade. +
                    Tuition.fees.up.to.date +
                    Unemployment.rate +
                    Inflation.rate +
                    GDP,
                  data = data, trace = FALSE)

cat("\n--- R sum  du Mod le (R f rence : Dropout) ---\n")
summary(model)
```

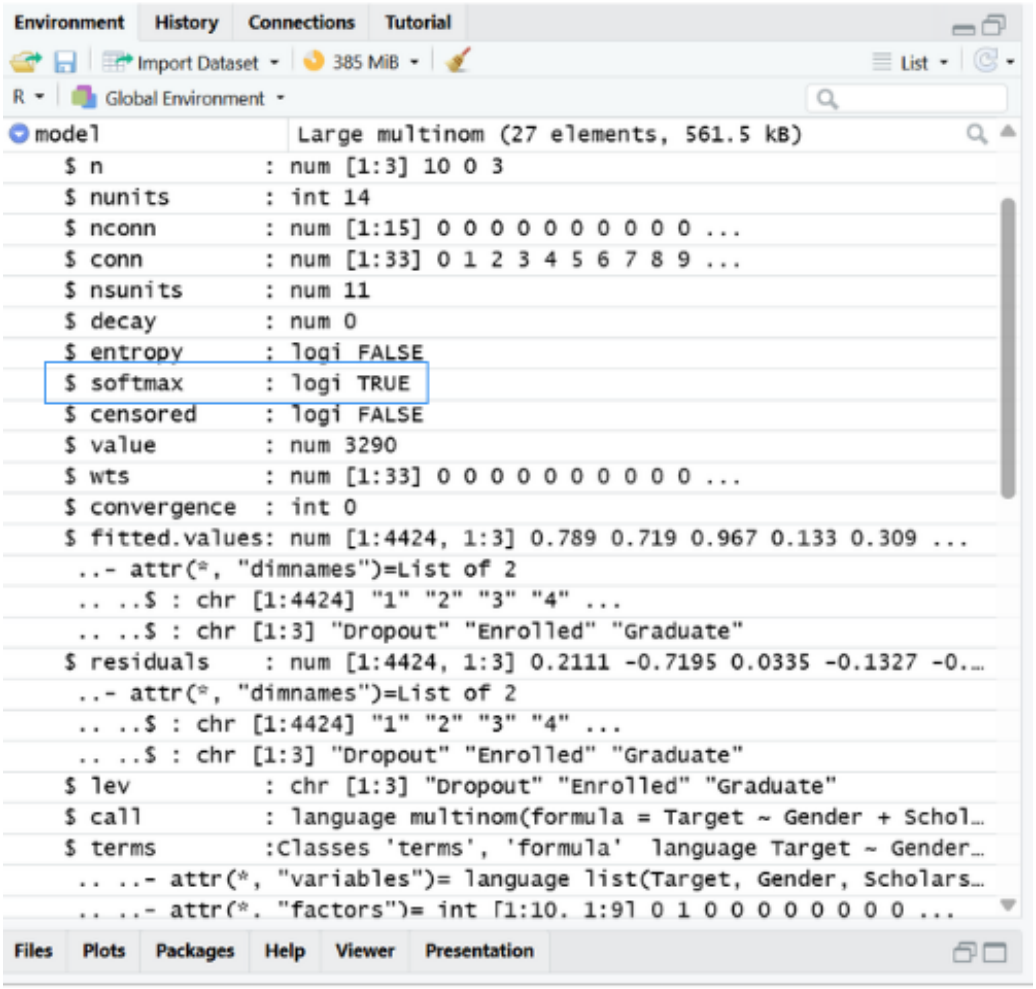

FIGURE 1 – Capture d'écran de l'exécution dans la console R

4 Analyse de la significativité et robustesse

```
# Calcul des p-values via les ratios de Wald
z <- summary(model)$coefficients / summary(model)$standard.errors
p_values <- (1 - pnorm(abs(z), 0, 1)) * 2

cat("\n--- P-Values des coefficients ---\n")
print(p_values)
```

```
Coefficients:
         (Intercept)    Gender1 Scholarship.holder1 Age.at.enrollment
Enrolled   -1.785014 -0.1519357           0.3290059       -0.04809784
Graduate   -4.606280 -0.6010367           1.3384404       -0.04558438
         Curricular.units.1st.sem..grade.
Enrolled                      -0.03260498
Graduate                       0.02821215
         Curricular.units.2nd.sem..grade. Tuition.fees.up.to.date1
Enrolled                        0.1856907                 1.939680
Graduate                        0.2839654                 3.260288
         Unemployment.rate Inflation.rate          GDP
Enrolled      -0.047955661    -0.03951319 0.0002231579
Graduate      -0.008933845    -0.02643016 0.0142484500

Std. Errors:
         (Intercept)    Gender1 Scholarship.holder1 Age.at.enrollment
Enrolled   0.3452228 0.10393840           0.1490684       0.006907824
Graduate   0.3682400 0.09729218           0.1307800       0.005916515
         Curricular.units.1st.sem..grade.
Enrolled                       0.01883225
Graduate                       0.02016298
         Curricular.units.2nd.sem..grade. Tuition.fees.up.to.date1
Enrolled                       0.01823745                0.1771586
Graduate                       0.01942641                0.2135477
         Unemployment.rate Inflation.rate        GDP
Enrolled        0.01980818     0.03682579 0.02396154
Graduate        0.01838703     0.03398431 0.02234402

Residual Deviance: 6580.173
AIC: 6620.173
>
```

FIGURE 2 – Capture d'écran du summary(model)

```
# Crit res d'Information pour la s lection de mod le
cat("\n--- Crit res de Robustesse ---\n")
cat("AIC :", AIC(model), "\n")
cat("BIC :", BIC(model), "\n")
```

```
> print(p_values)
          (Intercept)      Gender1 Scholarship.holder1 Age.at.enrollment
Enrolled 2.333256e-07 1.437998e-01          0.02730844      3.335554e-12
Graduate 0.000000e+00 6.506420e-10          0.00000000      1.310063e-14
         Curricular.units.1st.sem..grade.
Enrolled                        0.0833916
Graduate                        0.1617515
         Curricular.units.2nd.sem..grade. Tuition.fees.up.to.date1
Enrolled                                0                        0
Graduate                                0                        0
         Unemployment.rate Inflation.rate       GDP
Enrolled        0.01547775      0.2832819 0.9925693
Graduate        0.62705395      0.4367362 0.5236786
>
```

FIGURE 3 – Capture d'écran du summary(model)

5 Interprétation : Odds Ratios (RRR)

```
# Les coefficients exponenti s donnent les Relative Risk Ratios
rrr <- exp(coef(model))
cat("\n--- Relative Risk Ratios (Interpr tation par rapport au Dropout
    ) ---\n")
```

```r
print(rrr)
```

```
> rrr <- exp(coef(model))
> print(rrr)
         (Intercept)   Gender1 Scholarship.holder1 Age.at.enrollment
Enrolled 0.167794725 0.8590435            1.389586         0.9530405
Graduate 0.009988911 0.5482430            3.813092         0.9554390
         Curricular.units.1st.sem..grade.
Enrolled                        0.9679208
Graduate                        1.0286139
         Curricular.units.2nd.sem..grade. Tuition.fees.up.to.date1
Enrolled                         1.204050                 6.956521
Graduate                         1.328387                26.057049
         Unemployment.rate Inflation.rate      GDP
Enrolled         0.9531760      0.9612573 1.000223
Graduate         0.9911059      0.9739161 1.014350
> |
```

FIGURE 4 – Capture d'écran du summary(model)

6 Reproductibilité des résultats (Matrice de confusion)

```r
# Pr diction sur l' chantillon complet
predictions <- predict(model, newdata = data)
conf_matrix <- confusionMatrix(predictions, data$Target)

cat("\n--- Performance Globale du Mod le ---\n")
print(conf_matrix$table)
cat("\nPr cision (Accuracy) :", round(conf_matrix$overall['Accuracy']
    * 100, 2), "%\n")
```

```
> predictions <- predict(model, newdata = data)
> conf_matrix <- confusionMatrix(predictions, data$Target)
> print(conf_matrix$table)
          Reference
Prediction Dropout Enrolled Graduate
  Dropout      972      128      132
  Enrolled       8        6        1
  Graduate     441      660     2076
> cat("SCORE ACCURACY: ", round(conf_matrix$overall['Accuracy'] * 100, 2),
"%\n")
SCORE ACCURACY:  69.03 %
> cat("SCORE ACCURACY: ", round(conf_matrix$overall['Accuracy'] * 100, 2),
"%\n")
SCORE ACCURACY:  69.03 %
```

FIGURE 5 – Capture d'écran du summary(model)

7 Optimisation du modèle (Modèle Boosté)

Le modèle initial (69,03%) souffrait d'une confusion entre les classes Graduate et Enrolled. En intégrant le nombre d'unités validées (Approved) plutôt que simplement les notes, on capture mieux la progression réelle de l'étudiant.

```r
# Mod le optimis avec variables critiques
model_boosted <- multinom(Target ~
```

```r
                    Curricular.units.1st.sem..approved. +
                    Curricular.units.2nd.sem..approved. +
                    Curricular.units.2nd.sem..grade. +
                    Debtor +
                    Tuition.fees.up.to.date +
                    Scholarship.holder +
                    Age.at.enrollment +
                    Application.mode +
                    Course,
                  data = data, trace = FALSE)

# Nouvelle   valuation
new_preds <- predict(model_boosted, data)
new_conf <- confusionMatrix(new_preds, data$Target)

cat("Nouvelle Accuracy :", round(new_conf$overall['Accuracy'] * 100, 2)
    , "%\n")
```

```
> # Modèle optimisé avec variables critiques
> model_boosted <- multinom(Target ~
+                              # Performance pure
+                              Curricular.units.1st.sem..approved. +
+                              Curricular.units.2nd.sem..approved. +
+                              Curricular.units.2nd.sem..grade. +
+                              # Facteurs de blocage financier
+                              Debtor +
+                              Tuition.fees.up.to.date +
+                              Scholarship.holder +
+                              # Facteurs personnels
+                              Age.at.enrollment +
+                              Application.mode +
+                              Course,
+                           data = data, trace = FALSE)
>
> # Nouvelle évaluation
> new_preds <- predict(model_boosted, data)
> new_conf <- confusionMatrix(new_preds, data$Target)
>
> cat("Nouvelle Accuracy :", round(new_conf$overall['Accuracy'] * 100, 2),
"%\n")
Nouvelle Accuracy : 73.17 %
>
```

FIGURE 6 – Capture d'écran du summary(model)

FIN DU CODE R - DOSSIER 4